Unit: 6 Unit 2: Exponents Radicals
Topic: Exponents

Name: ________________
Date: ____________

Foundation (Jalapeno)

Q1. What is the value of $2^3 \cdot 2^4$?
(A) 32
(B) 64
(C) 128
(D) 256
(E) 512

Q2. Simplify $(3^2)^3$ to get
(A) 3^5
(B) 3^6
(C) 3^8
(D) 3^9
(E) 3^{10}

Q3. A space probe travels 2^5 miles in one day. How many miles does it travel in 3 days?
(A) 2^{15}
(B) 2^8
(C) $3 \cdot 2^5$
(D) $2^{15} - 2^5$
(E) 2^{10}

Q4. Simplify $\frac{2^8}{2^3}$ to get
(A) 2^2
(B) 2^5
(C) 2^{11}
(D) 2^{10}
(E) 2^6

Q5. A laboratory has 3^4 test tubes. If they want to package them in boxes of 3^2 tubes each, how many boxes can they make?
(A) 9
(B) 27
(C) 81
(D) 3^2
(E) 3^{4-2}

Q6. The mass of a planet is 2^{10} kg. If its mass increases by a factor of 2^2, what is its new mass?
(A) $2^{10} + 2^2$
(B) $2^{10} \cdot 2^2$
(C) $2^{12} - 2^{10}$
(D) $2^{10}/2^2$
(E) $2^{10} - 2^2$

Q7. What is the simplified form of $(2^3)^2 \cdot 2^4$?
(A) 2^{10}
(B) 2^{12}
(C) 2^{14}
(D) 2^{16}
(E) 2^{18}

Q8. The distance between two stars is 3^5 light-years. If a spaceship travels 3^3 light-years per day, how many days will it take to travel between the two stars?
(A) 3^2
(B) $3^5/3^3$
(C) 3^{5-3}
(D) 3^{5+3}
(E) $3^{5\cdot 3}$

Open Ended Questions (FRQ)

Q9. A rocket's altitude above the Earth's surface is given by the equation $h(t) = 2^{t+1}$, where h is the altitude in kilometers and t is the time in hours. Find the altitude after 4 hours.

Q10. The volume of a sphere with radius r is given by the equation $V = \frac{4}{3}\pi r^3$. If the radius of the sphere is 2^2 meters, find the volume of the sphere.

Chef's Tips

- Provide clear instructions and examples before the test
- Ensure students have the necessary resources and materials
- Be available to answer questions and provide guidance during the test